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Name: _____

July 4, 2015, Saturday

Department/School: _____

SID: _____

University Physics I
Final Examination

Kuang Yaming Honors School, Nanjing University

Gamma function: $\Gamma(\nu) = \int_0^\infty e^{-t} t^{\nu-1} dt$, $\Gamma(\nu+1) = \nu\Gamma(\nu)$, $\Gamma(1) = \Gamma(2) = 1$, $\Gamma(1/2) = \sqrt{\pi}$.

Gauss integral: $I_n = \int_0^\infty x^n e^{-x^2} dx = \frac{1}{2} \Gamma\left(\frac{n+1}{2}\right)$, $I_{2m-1} = \frac{1}{2}(m-1)!$, $I_{2m} = \frac{(2m-1)!!}{2^{m+1}} \sqrt{\pi}$.

Select five out of following six problems.

1.(20pts) An experimenter arranges to trigger two flashbulbs simultaneously, producing a big flash located at the origin of his reference frame and a small flash at $x = 30.0 \text{ km}$. An observer moving at a speed of $0.250c$ in the positive direction of x -axis also views the flashes.

(a) What is the time interval between them according to the observer?

(b) Which flash does the observer say occurs first?

Solution:(a) In the reference frame of the experimenter, the space and time intervals of two events are

$$\Delta x = x_2 - x_1 = 30.0 \text{ km}, \quad \Delta t = t_2 - t_1 = 0.$$

Using the Lorentz transformation, in the observer's frame, we have

$$\begin{aligned} \Delta t' &= t'_2 - t'_1 = \gamma \left(\Delta t - \frac{\beta}{c} \Delta x \right) = -\gamma \frac{\beta}{c} \Delta x = -\frac{\beta}{\sqrt{1-\beta^2}} \frac{\Delta x}{c} \\ &= -\frac{0.25}{\sqrt{1-0.25^2}} \frac{30.0 \times 10^3 \text{ m}}{3.0 \times 10^8 \text{ m/s}} = -\frac{1}{\sqrt{15}} \times 10^3 \text{ s} = -0.258 \times 10^3 \text{ s}. \end{aligned}$$

(2) In the observer's frame, $t'_2 < t'_1$. The small flash occurs firstly.

2.(20 pts) Galaxy A is reported to be receding from us with a speed of $0.35c$. Galaxy B, located in precisely the opposite direction, is also found to be receding from us at this same speed.

(a) What recessional speed would an observer on Galaxy A find for our galaxy?

(b) What recessional speed would an observer on Galaxy A find for Galaxy B?

Solution: (a) By taking the reference frame of our galaxy as frame S , the velocity of our galaxy in S is

$$v_x = v_y = v_z = 0.$$

We take the reference frame the Galaxy A as reference frame S' . Since Galaxy A is receding from us with a speed of $0.35c$, then $u = 0.35c$. By using the Lorentz transformation of velocity, the velocity of our galaxy in S' is

$$\begin{aligned} v'_x &= \frac{v_x - u}{1 - \beta v_x / c} = -u = -0.35c, \\ v'_y &= \frac{v_y}{\gamma(1 - \beta v_x / c)} = 0, \\ v'_z &= \frac{v_z}{\gamma(1 - \beta v_x / c)} = 0. \end{aligned}$$

Therefore, the recessional speed an observer on Galaxy A would find for our galaxy is

$$v' = \sqrt{v'^2_x + v'^2_y + v'^2_z} = 0.35c.$$

(b) In frame S , the velocity of Galaxy B is

$$v_x = -u = -0.35c, \quad v_y = v_z = 0.$$

By using the Lorentz transformation of velocity, the velocity of Galaxy B in S' is

$$v'_x = \frac{v_x - u}{1 - \beta v_x / c} = \frac{-2u}{1 + \beta u / c} = -2 \frac{\beta}{1 + \beta^2} c = -2 \frac{0.35}{1 + 0.35^2} c = -0.62c,$$

$$v'_y = \frac{v_y}{\gamma(1 - \beta v_x / c)} = 0,$$

$$v'_z = \frac{v_z}{\gamma(1 - \beta v_x / c)} = 0.$$

Therefore, the recessional speed an observer on Galaxy A would find for Galaxy B is

$$v' = \sqrt{v_x'^2 + v_y'^2 + v_z'^2} = 0.62c.$$

3.(20pts) Under the standard pressure of atmosphere, 10 kg of water at 20°C is turned into the steam at 250°C . The specific heats at constant pressure for the water and steam are $4187\text{ J/kg}\cdot\text{K}$ and $1670\text{ J/kg}\cdot\text{K}$, respectively. The vaporizing heat for water is $2.25 \times 10^6\text{ J/kg}$. Find the change in entropy of the water ΔS .

Solution: The change in entropy for the liquid water to be heated from 20°C to 100°C is

$$\begin{aligned}\Delta S_1 &= \int_{(273.15+20)\text{K}}^{(273.15+100)\text{K}} mc_p^l \frac{dT}{T} = mc_p^l \int_{293.15\text{K}}^{373.15\text{K}} \frac{dT}{T} \\ &= m_w c_p^l \ln\left(\frac{373.15}{293.15}\right) = 10\text{ kg} \times 4187\text{ J/kg}\cdot\text{K} \times \ln\left(\frac{373.15}{293.15}\right) \\ &= 0.100 \times 10^5\text{ J/K}.\end{aligned}$$

The change in entropy for the transition of the liquid water of 100°C into the water steam of 100°C is

$$\Delta S_2 = \frac{\Delta Q}{T_c} = \frac{mL}{T_c} = \frac{10\text{ kg} \times 2.25 \times 10^6\text{ J/kg}}{373.15\text{K}} = 0.603 \times 10^5\text{ J/K}.$$

The change in entropy for the water steam of 100°C to be heated to 250°C is

$$\begin{aligned}\Delta S_3 &= \int_{(273.15+100)\text{K}}^{(273.15+250)\text{K}} mc_p^s \frac{dT}{T} = mc_p^s \int_{373.15\text{K}}^{523.15\text{K}} \frac{dT}{T} \\ &= mc_p^s \ln\left(\frac{523.15}{373.15}\right) = 10\text{ kg} \times 1670\text{ J/kg}\cdot\text{K} \times \ln\left(\frac{523.15}{373.15}\right) \\ &= 0.561 \times 10^4\text{ J/K}.\end{aligned}$$

Therefore, the total change in entropy is

$$\Delta S = \Delta S_1 + \Delta S_2 + \Delta S_3 = 0.759 \times 10^5\text{ J/K}.$$

4.(20pts) Suppose that the molar heat capacity at constant volume for a monatomic ideal gas is $c_v = 3R/2$, where R is the gas constant, and the molar internal energy u of it is not dependent on its volume, i.e., $(\partial u / \partial v)_T = 0$, where v is the molar volume of the gas.

(a) Show that $du = \frac{3}{2} R dT$.

(b) Show that the differential of molar entropy s of the gas is given by $ds = c_p \frac{dT}{T} - R dp$, where $c_p = \frac{5}{2} R$ is the molar heat capacity at constant pressure of the gas.

(c) Show further that $ds = \frac{3}{2} R \frac{dp}{p} + \frac{5}{2} R \frac{dv}{v}$.

(d) Show that $\kappa_s = \frac{3}{5} \kappa_T$ where $\kappa_s = -\frac{1}{v} \left(\frac{\partial v}{\partial p} \right)_s$ and $\kappa_T = -\frac{1}{v} \left(\frac{\partial v}{\partial p} \right)_T$.

Solution: (a) For the ideal gas $\left(\frac{\partial u}{\partial v} \right)_T = 0$. Then

$$du = \left(\frac{\partial u}{\partial T} \right)_v dT + \left(\frac{\partial u}{\partial v} \right)_T dv = c_v dT = \frac{3}{2} R dT. \quad (1)$$

(b) The fundamental equation of thermodynamics is $du = T ds - p dv$. Using the equation of state $pv = RT$ and (1),

we have

$$ds = \frac{du}{T} + \frac{p}{T} dv = \frac{3R}{2} \frac{dT}{T} + R \frac{dv}{v} \quad (2)$$

Making differential of both sides of $\ln p + \ln v = \ln T + \ln R$, we have

$$\frac{dp}{p} + \frac{dv}{v} = \frac{dT}{T} \quad \text{or} \quad \frac{dv}{v} = \frac{dT}{T} - \frac{dp}{p}. \quad (3)$$

Substituting above equation into (2), we obtain

$$ds = \frac{3R}{2} \frac{dT}{T} + R \frac{dv}{v} = \frac{5R}{2} \frac{dT}{T} - R \frac{dp}{p} = c_p \frac{dT}{T} - R \frac{dp}{p},$$

where

$$c_p = T \left(\frac{\partial s}{\partial T} \right)_p = \frac{5R}{2}$$

is the isobaric molar heat capacity of the gas.

(c) Substituting the first equation of (3) into (2), we have

$$ds = \frac{3R}{2} \frac{dT}{T} + R \frac{dv}{v} = \frac{3R}{2} \left(\frac{dp}{p} + \frac{dv}{v} \right) + R \frac{dv}{v} = \frac{3R}{2} \frac{dp}{p} + \frac{5R}{2} \frac{dv}{v}. \quad (4)$$

(d) Rewriting the equation of state as $\ln v = \ln T - \ln p + \ln R$, and taking partial derivatives with respect to temperature p , we have

$$\left(\frac{\partial \ln v}{\partial p} \right)_T = -\frac{1}{p}.$$

$$\text{Thus } \kappa_T = -\frac{1}{v} \left(\frac{\partial v}{\partial p} \right)_T = -\left(\frac{\partial \ln v}{\partial p} \right)_T = \frac{1}{p}.$$

Eq. (4) can be rewritten as

$$\frac{dv}{v} = \frac{2}{5R} ds - \frac{3}{5} \frac{dp}{p}.$$

For $ds = 0$, $\frac{dv}{v} = -\frac{3}{5} \frac{dp}{p}$. Thus, we have

$$\kappa_S = -\frac{1}{v} \left(\frac{\partial v}{\partial p} \right)_s = \frac{3}{5} \frac{1}{p}.$$

Therefore $\kappa_S = \frac{3}{5} \kappa_T$.

5.(20pts) Consider a gas with a fixed number of particles. The fundamental equation of thermodynamics of it is given by $dU = TdS - pdV$.

(a) Show that $dF = -SdT - pdV$ where $F = U - TS$ is the Helmholtz free energy of the system, and that

$$\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial p}{\partial T} \right)_V.$$

(b) Show that $dS = \frac{C_V}{T} dT + \left(\frac{\partial p}{\partial T} \right)_V dV$, where C_V is the heat capacity of the gas at constant volume.

(c) Show that $dU = C_V dT + \left[T \left(\frac{\partial p}{\partial T} \right)_V - p \right] dV$.

(d) Show that Joule coefficient $K = \left(\frac{\partial T}{\partial V} \right)_U \neq 0$ for the van der Waals gas with one mole molecules.

(Hint: The van der Waals equation for one mole of molecules can be written as $p = \frac{RT}{V-b} - \frac{a}{V^2}$)

Solution: (a) $dF = d(U - TS) = dU - TdS - SdT = -SdT - pdV$.

Since dF is an exact (a total) differential, therefore

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V.$$

(b) and (c)
From

$$dU = TdS - pdV = T\left(\frac{\partial S}{\partial T}\right)_V dT + \left[T\left(\frac{\partial S}{\partial V}\right)_T - p\right]dV = T\left(\frac{\partial S}{\partial T}\right)_V dT + \left[T\left(\frac{\partial p}{\partial T}\right)_V - p\right]dV.$$

we have $C_V = \left(\frac{\partial U}{\partial T}\right)_V = T\left(\frac{\partial S}{\partial T}\right)_V.$

Thus

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV = \frac{C_V}{T} dT + \left(\frac{\partial p}{\partial T}\right)_V dV,$$

$$dU = C_V dT + \left[T\left(\frac{\partial p}{\partial T}\right)_V - p\right]dV.$$

(d) For $dU = 0$, we have

$$C_V dT + \left[T\left(\frac{\partial p}{\partial T}\right)_V - p\right]dV = 0.$$

Thus

$$K = \left(\frac{\partial T}{\partial V}\right)_U = \frac{1}{C_V} \left[p - T\left(\frac{\partial p}{\partial T}\right)_V \right].$$

For a mole of van der Waals gas,

$$\left(\frac{\partial p}{\partial T}\right)_V = \frac{R}{V-b}, \quad p - T\left(\frac{\partial p}{\partial T}\right)_V = -\frac{a}{V^2}.$$

Therefore

$$K = \left(\frac{\partial T}{\partial V}\right)_U = -\frac{a}{C_V V^2} \neq 0.$$

6.(20pts) The ideal gas of total number of molecules N obeys the Maxwell velocity distribution

$$dn = C \exp\left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right] dv_x dv_y dv_z$$

where m is the mass of the molecule, T the temperature, and k_B the Boltzmann constant. And the multiplier C is to be determined.

(a) Determine the multiplier C in terms of m , k_B , T and N .

(b) Find $\langle v_x \rangle$ and $\langle v_y \rangle$, the averages of the velocity components v_x and v_y .

(c) Find $\langle v_x^2 \rangle$, $\langle v_y^2 \rangle$ and $\langle v_x v_y \rangle$.

(d) Find the fluctuation of the x -component of velocity $\Delta v_x \equiv \sqrt{\langle (v_x - \langle v_x \rangle)^2 \rangle}$.

Solution:

$$\begin{aligned} N &= \int dn \\ &= C \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right] dv_x dv_y dv_z \\ &= C \left(\int_{-\infty}^{\infty} e^{-\frac{mv_x^2}{2k_B T}} dv_x \right)^3. \end{aligned}$$

Since

$$\int_{-\infty}^{\infty} e^{-\frac{mv_x^2}{2k_B T}} dv_x = \sqrt{\frac{2k_B T}{m}} \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\frac{2k_B T}{m}} \sqrt{\pi} = \sqrt{\frac{2\pi k_B T}{m}},$$

we have

$$C = N \left(\frac{m}{2\pi k_B T} \right)^{\frac{3}{2}}.$$

(b)

$$\begin{aligned} \langle v_x \rangle &= \frac{\int v_x dn}{\int dn} = \frac{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} v_x \exp \left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T} \right] dv_x dv_y dv_z}{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp \left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T} \right] dv_x dv_y dv_z} \\ &= \frac{\int_{-\infty}^{\infty} v_x \exp \left(-\frac{mv_x^2}{2k_B T} \right) dv_x}{\int_{-\infty}^{\infty} \exp \left(-\frac{mv_x^2}{2k_B T} \right) dv_x} = 0. \end{aligned}$$

(The numerator of above expression is the integral of an odd function from negative infinity to positive infinity and is therefore zero.)

Similarly

$$\langle v_y \rangle = 0.$$

(c)

$$\begin{aligned} \langle v_x^2 \rangle &= \frac{\int v_x^2 dn}{\int dn} = \frac{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} v_x^2 \exp \left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T} \right] dv_x dv_y dv_z}{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp \left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T} \right] dv_x dv_y dv_z} \\ &= \frac{\int_{-\infty}^{\infty} v_x^2 \exp \left(-\frac{mv_x^2}{2k_B T} \right) dv_x}{\int_{-\infty}^{\infty} \exp \left(-\frac{mv_x^2}{2k_B T} \right) dv_x} = \frac{2 \int_0^{\infty} v_x^2 \exp \left(-\frac{mv_x^2}{2k_B T} \right) dv_x}{2 \int_0^{\infty} \exp \left(-\frac{mv_x^2}{2k_B T} \right) dv_x} \\ &= \frac{2k_B T}{m} \frac{\int_0^{\infty} t e^{-t} \frac{dt}{2\sqrt{t}}}{\int_0^{\infty} e^{-t} \frac{dt}{2\sqrt{t}}} = \frac{2k_B T}{m} \frac{\int_0^{\infty} t^{\frac{1}{2}} e^{-t} dt}{\int_0^{\infty} t^{-\frac{1}{2}} e^{-t} dt} = \frac{2k_B T}{m} \frac{\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{1}{2}\right)} = \frac{k_B T}{m}. \end{aligned}$$

($t = \frac{m}{2k_B T} v_x^2$ or $v_x = \sqrt{\frac{2k_B T}{m}} \sqrt{t}$ is used in the above derivation.)

Similarly

$$\langle v_y^2 \rangle = \frac{k_B T}{m}.$$

$$\begin{aligned}
\langle v_x v_y \rangle &= \frac{\int v_x v_y dn}{\int dn} = \frac{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} v_x v_y \exp\left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right] dv_x dv_y dv_z}{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left[-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right] dv_x dv_y dv_z} \\
&= \frac{\int_{-\infty}^{\infty} v_x \exp\left(-\frac{mv_x^2}{2k_B T}\right) dv_x \int_{-\infty}^{\infty} v_y \exp\left(-\frac{mv_y^2}{2k_B T}\right) dv_y}{\int_{-\infty}^{\infty} \exp\left(-\frac{mv_x^2}{2k_B T}\right) dv_x \int_{-\infty}^{\infty} \exp\left(-\frac{mv_y^2}{2k_B T}\right) dv_y} = 0.
\end{aligned}$$

(d)

$$\Delta v_x \equiv \sqrt{\langle (v_x - \langle v_x \rangle)^2 \rangle} = \sqrt{\langle v_x^2 \rangle - \langle v_x \rangle^2} = \sqrt{\frac{k_B T}{m}}.$$